

KANGAROO MOTHER CARE SIMULATOR

NeoNiu

To improve the wellbeing of premature newborns

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Social Impact



13,4M

**Premature births
worldwide/year**

Global rate: 9.9%
(range 4–16% by country)
WHO/UNICEF, 2023



19.495

**Premature births
in Spain**

6.13% of births
(INE, 2024)
88% moderate-to-late preterm



3.396

**Premature births
in Catalonia**

6.44% of births
(INE, 2024)
4.1% extremely preterm

REQUIREMENTS for the Simulator

Technical and functional
features of the device



Play soothing sounds
for the newborn



Simulate the chest movement
of the parents



Safe for the newborn



Non-intrusive — allows
diagnostic imaging



Comfortable — suitable
materials for the newborn



Silent — maximum
calm in the NICU



Compact — does not interfere
with clinical staff

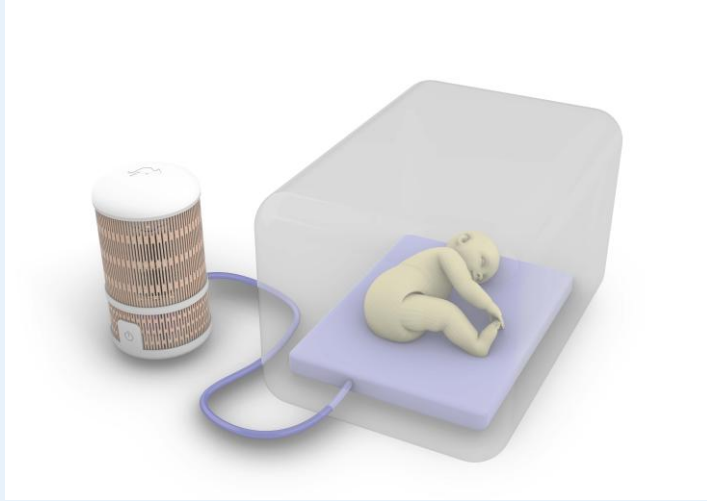


Simple — easy
maintenance



Affordable — accessible
to any hospital

Proposed Solution



[Device illustration]

Air Mattress

Elastic resonance chamber inside the incubator. No metallic or electromagnetic components.

Control Unit

Generates the air pulse and plays sounds. Located outside the incubator.

Transmission Tube

Connects both elements forming a closed air circuit. Transmits sound and movement.

Prototype Contributions



Mimics Kangaroo Mother Care

Simulates contact when physical contact between newborn and parent is not possible.



Physiological Movement

Adjusts frequency and intensity to match adult breathing (12–20 breaths/min).



No Electronics Near the Newborn

No electronic, metallic or electromagnetic component near the incubator.



Sound Transmission

Sound travels through the tube. Only audible with the ear resting on the air mattress.



Personalised Sounds

Plays heartbeat, breathing and parental voice recordings (configurable MP3 files).



Non-Intrusive

Compatible with diagnostic imaging without moving or disconnecting the newborn.

HOW IT WORKS

Inspiration and
expiration cycle



DEVICE STATUS

STATUS	SOUND	AIR FLOW
Off	Muted	Stopped
Active	Active	Inspiration / Expiration

ACKNOWLEDGEMENTS

NICU Staff — Hospital Vall d'Hebron, Barcelona

For caring for my children and helping the NeoNiu idea take root.

Staff of the Ateneu de Fabricació de les Corts

For all their support and contributions to the prototype development.

Friends and family

For supporting me through the challenges of this project.